

About SIMULOS®

Simulation Governance Architecture

Reality is increasingly simulated before it is entered.

Systems are tested before they are built.

Autonomous systems are exposed to conditions before they encounter them.

Missions are evaluated before they are launched.

Infrastructure is stressed before it fails.

Future conditions are explored before they exist.

Simulation is becoming part of how organizations understand, design, test, compare, prepare, validate, and make decisions about reality.

SIMULOS® — Simulation Governance Architecture was created to provide a stable governance architecture for this expanding simulation domain.

SIMULOS® does not define simulation software, numerical solvers, physics engines, digital-twin platforms, modeling languages, computing infrastructure, or implementation technologies.

SIMULOS® governs simulation. Why SIMULOS® Exists

Simulation can generate enormous amounts of information.

But the existence of simulation does not itself answer fundamental governance questions:

What exactly is being simulated?

Why is it being simulated?

Which parts of reality are represented?

Which assumptions make that representation possible?

Which models, environments, inputs, and scenarios define the simulation?

How was the simulation executed?

What do its outputs actually mean?

How closely do those outputs correspond to reality?

What may simulation-derived evidence legitimately support?

When does changing reality require resimulation?

SIMULOS® addresses these questions by defining the governable spaces that exist throughout the complete simulation lifecycle.

Rather than standardizing one simulation technology, SIMULOS® standardizes the governance conditions under which simulation can be identified, scoped, represented, executed, interpreted, correlated with reality, converted into bounded evidence, relied upon, monitored, and resimulated. Simulation Before Reliance

The objective of SIMULOS® is not to prescribe how a simulation must be technically built.

The objective is to define how simulation must remain governable.

Every simulation has an object.

Every Simulation Object requires a purpose and scope.

Every scope determines what reality must be represented.

Every representation depends upon explicit or implicit assumptions.

Every simulation requires a model and environment.

Every model operates upon inputs and data.

Every meaningful simulation requires relevant scenarios.

Every scenario must be executed under governed conditions.

Every execution produces outputs.

Every output requires interpretation before broader meaning can be attributed to it.

Claims about reality require a governed simulation-to-reality relationship.

Simulation-derived evidence requires qualification before reliance.

Continuing reliance requires lifecycle monitoring and, where material change occurs, resimulation.

SIMULOS® therefore treats simulation as an independent governance domain. The Simulation Governance Architecture

SIMULOS® consists of twelve independent simulation governance spaces.

Each space answers one architectural question.

Each space governs one distinct part of the simulation lifecycle.

Each space has a defined beginning, operational purpose, and governance boundary.

Each space begins where the preceding simulation governance requirement has been sufficiently established.

Together, these spaces form the complete Simulation Governance Architecture.

The architecture progresses through:

Simulation Object → Simulation Purpose & Scope → Simulation Reality
Representation → Simulation Assumptions → Simulation Model & Environment →
Simulation Inputs & Data → Simulation Scenario Design → Simulation Execution →
Simulation Output Governance → Simulation-to-Reality Correlation → Simulation
Evidence & Reliance Boundary → Simulation Lifecycle, Drift & Resimulation

This progression represents the governance journey from identifying what is being simulated to determining whether simulation-derived evidence remains legitimate enough to support continuing reliance as reality changes .
Simulation Spaces Rather Than Technical Components

SIMULOS® does not organize simulation governance around software packages, simulation engines, computing architectures, vendors, departments, or technical components.

Instead, it organizes governance around distinct simulation governance spaces.

Each space represents an independent governance responsibility that exists regardless of the simulation technology used.

A digital twin, autonomous-driving simulator, spacecraft mission simulator, engineering model, scientific simulation, physical test simulation, AI training environment, or future simulation technology may be technically very different.

Yet each may still require governance of:

object → purpose → representation → assumptions → model → inputs → scenarios → execution → outputs → reality correlation → evidence → lifecycle.

Because these spaces are technology-neutral, SIMULOS® can remain applicable while simulation technologies continue to evolve. A Methodology Rather Than an Implementation

SIMULOS® is a governance methodology.

It is not a simulation implementation framework.

The architecture defines which simulation governance conditions must exist, how they relate to one another, where their boundaries lie, and what must remain explicit before simulation results acquire broader meaning or reliance.

Organizations remain free to select their own: simulation platforms, modeling methods, numerical methods, physics engines, digital-twin technologies, AI models, computing infrastructure, scenario-generation systems, verification tools, data systems, engineering processes.

SIMULOS® governs the methodological structure surrounding those implementations rather than replacing them.

This separation allows the architecture to remain stable while simulation technology changes. From Simulation Object to Governed Evidence

Simulation cannot be governed as a single computational event.

Before a simulation can be scoped, the Simulation Object must be identified.

Before reality can be represented, the purpose and scope must be known.

Before models can legitimately represent reality, assumptions and representation boundaries must be explicit.

Before execution, models, environments, inputs, data, and scenarios must be sufficiently governed.

Before outputs can become evidence, their meaning must be established.

Before simulation-derived evidence can support broader claims, the relationship between simulation and reality must be understood.

Before evidence can support reliance, its qualification, weight, applicability, uncertainty, and limitations must be bounded.

And before historical simulation evidence can continue to be relied upon, the simulation basis must remain relevant as reality changes.

SIMULOS® transforms these dependencies into a structured and navigable simulation governance lifecycle. Simulation Is Not Reality

One of the most important distinctions within SIMULOS® is:

Simulation ≠ Reality

A simulation may reproduce aspects of reality with extraordinary precision.

It may execute billions of scenarios.

It may demonstrate behavior that appears stable across enormous parameter spaces.

It may correlate strongly with observed reality.

None of these facts alone makes the simulation equivalent to reality.

SIMULOS® therefore preserves the distinction between:

Represented Reality

and

Operational Reality.

This distinction enables simulation to be used powerfully without allowing computational representation to acquire greater authority than its governed basis can support. Simulation Output Is Not Automatically Evidence

SIMULOS® also establishes another fundamental distinction:

Simulation Output ≠ Simulation Evidence

A simulation engine may produce millions or billions of outputs.

Those outputs do not automatically become evidence merely because they were computationally generated.

Simulation-derived information must first acquire a defined evidential purpose, remain traceable to its simulation basis, and retain the assumptions, scenarios, conditions, correlations, uncertainties, and limitations that determine its legitimate meaning.

Only then can simulation-derived information enter governed evidential reliance.

This creates a methodological boundary between:

what a simulation produced

and

what an organization may legitimately conclude from what the simulation produced. Simulation Evidence Is Not Validation

SIMULOS® deliberately separates simulation evidence governance from broader validation governance.

SIMULOS® determines what simulation-derived evidence can legitimately mean.

VALIDOS® — Validation Governance Architecture determines whether the complete evidence basis is sufficient to support a Validation Determination and intended reliance.

This distinction prevents simulation success from automatically becoming a claim of real-world validity.

A simulation may provide strong evidence.

It may provide decisive evidence for a narrowly defined question.

It may substantially reduce uncertainty.

But simulation-derived evidence remains one governed evidential source whose broader validation significance must be determined within the appropriate validation architecture. The Simulation-to-Reality Boundary

The transition from simulation to reality is one of the most consequential points in the architecture.

A simulated autonomous vehicle may perform successfully across millions of scenarios.

A simulated spacecraft may survive thousands of modeled failure conditions.

A digital twin may reproduce observed industrial behavior with high accuracy.

A scientific model may repeatedly predict known outcomes.

These results matter.

But the governance question is not simply:

Did the simulation succeed?

The stronger question is:

What relationship has actually been established between the simulated result and the reality to which that result is being applied?

SIMULOS® therefore treats Simulation-to-Reality Correlation as an independent governable space rather than assuming that simulation performance automatically transfers into operational reality. Lifecycle Governance and Resimulation

Simulation does not end when execution stops.

Reality changes.

Simulation Objects change.

Models change.

Inputs change.

Assumptions become obsolete.

Operating environments evolve.

New evidence appears.

Previously unknown scenarios become relevant.

A simulation that legitimately supported a conclusion yesterday may no longer provide the same evidential basis tomorrow.

SIMULOS® therefore extends governance beyond execution and evidence into Simulation Lifecycle, Drift & Resimulation.

Where material change affects the simulation basis, the architecture supports governed re-entry into the earliest materially affected simulation space.

The result is not uncontrolled repetition of the complete lifecycle.

It is targeted governed resimulation. Who May Benefit

SIMULOS® may support: AI developers and operators, autonomous-system developers, robotics organizations, digital-twin environments, engineering organizations, scientific institutions, research laboratories, aerospace and space organizations, automotive and transportation organizations, healthcare and medical simulation environments, financial and economic modeling environments, industrial operators, infrastructure planners, safety and reliability teams, public institutions, regulators, assurance and validation organizations, critical infrastructure providers, organizations using simulation to support material decisions.

Relationship to Other OOF® Architectures

SIMULOS® governs simulation.

It operates alongside complementary OOF® architectures.

GOA™ — Governance Architecture governs governance structure, authority, scope, delegation, accountability, continuity, and interoperability.

ORA™ — Operational Reality Architecture governs Operational Reality and provides the methodological distinction between what actually exists or occurs and what is represented through simulation.

VALIDOS® — Validation Governance Architecture governs Validation Objects, validation purpose and scope, criteria, methods, evidence fitness, source reliability, execution, determination, Validation States, reliance, and revalidation.

INTEGROS® — Integrity Governance Architecture governs integrity across the complete integrity lifecycle.

OBIDENTITY® — Origin-Bound Identity Governance Architecture governs origin, identity, ownership, authority, permission, provenance, transfer, continuity, and auditability.

AGA™ — Accountability Governance Architecture governs accountability, responsibility, attribution, delegation, and accountable relationships.

AIG® — AI Governance Architecture governs AI behavior.

ASGA™ — Autonomous Systems Governance Architecture governs autonomous systems, autonomous agents, autonomous authority, constraints, execution, and operational autonomy.

CLIA® — Cognitive Governance Intelligence Architecture governs cognition, reasoning, interpretation, and cognitive integrity.

MGIA™ — Memory Governance Intelligence Architecture governs memory, memory continuity, and memory integrity.

ADIT® — Audit in Real Time Architecture governs continuous auditability and evidential observation across operational lifecycles.

Together, these architectures allow simulation to interact with operational reality, validation, integrity, accountability, AI, autonomy, cognition, memory, and auditability without collapsing those independent governance domains into one architecture. Use Case 1 — Autonomous System Before Real-World Deployment

An organization develops an autonomous robot intended to operate in complex physical environments.

Before deployment, the robot is exposed to millions of simulated scenarios involving obstacles, people, environmental variation, sensor degradation, unusual events, system failures, and behavioral edge cases.

A conventional analysis may report:

99.97% simulation success.

SIMULOS® asks a much larger set of questions.

What robot configuration was actually simulated?

Which operational reality was represented?

Which physical effects were simplified?

Which assumptions were required?

Were the inputs representative?

Which scenarios were absent?

Were repeated executions genuinely diverse?

What did the outputs actually demonstrate?

How strongly did simulated behavior correlate with physical tests?

What proposition can the resulting evidence legitimately support?

Where does reliance upon that evidence stop?

The result may be that simulation provides substantial evidence for performance across the represented scenario space while still requiring physical evidence before unrestricted operational deployment.

SIMULOS® does not make simulation less useful.

It makes the meaning of simulation success governable. Use Case 2 — Future Space Mission Where Complete Reality Does Not Yet Exist

A space organization is preparing a mission to an environment that cannot be completely reproduced on Earth and has never been experienced in the exact intended mission configuration.

Simulation becomes essential.

Spacecraft behavior, navigation, landing, environmental interaction, equipment failure, mission contingencies, crew behavior, communication delays, and emergency conditions may all be simulated before the mission exists as Operational Reality.

SIMULOS® provides the governance structure for establishing:

what is being simulated;

why;

which aspects of the future environment are represented;

which assumptions substitute for unavailable knowledge;

which models and data create the representation;

which scenarios must be explored;

how executions are controlled;

what the resulting outputs mean;

which parts can be correlated with existing reality;

what evidential weight those results may legitimately carry;

and which assumptions or reality changes would require resimulation.

The simulation can therefore provide highly valuable evidence without being represented as proof of conditions that have never yet existed.

Where reality cannot yet be fully observed, simulation may extend knowledge—but governance must define where that knowledge ends. The Long-Term Vision

Simulation is moving from a specialized engineering technique toward a foundational infrastructure for understanding and acting upon reality.

AI systems may learn inside simulated worlds.

Autonomous systems may experience millions of synthetic situations before encountering physical ones.

Digital twins may continuously mirror infrastructure, factories, cities, biological systems, economies, and operational environments.

Scientific simulations may explore conditions humans cannot directly observe.

Organizations may increasingly make consequential decisions about real systems based upon events that occurred only inside simulated environments.

As this transition accelerates, the central question will no longer be only whether organizations can build more powerful simulations.

It will also be:

How can society know what those simulations legitimately mean?

The long-term objective of SIMULOS® is to provide a common simulation governance language through which organizations can identify, structure, compare, govern, evaluate, rely upon, monitor, and resimulate representations of reality without confusing simulation capability with reality itself.

SIMULOS® is intended to make simulation governable before execution, during execution, after execution, at the boundary with reality, at the boundary with evidence, and throughout continuing reliance. Canonical Principle

Simulation technologies, models, environments, computational capabilities, and represented realities will continue to change. Simulation governance must remain stable. SIMULOS® therefore governs the methodological conditions that determine what is being simulated, why it is being simulated, how reality is represented, what simulation results legitimately mean, how those results relate to reality, how far simulation-derived evidence may be relied upon, and when material change requires governed resimulation. Canonical Closing Statement

SIMULOS® — Simulation Governance Architecture provides the governance foundation through which simulation can be formally identified, purpose-bounded, reality-represented, assumption-governed, model-structured, input-controlled, scenario-designed, execution-governed, output-interpreted, reality-correlated, evidentially bounded, and continuously maintained through lifecycle governance and resimulation. By defining the independent governable spaces of the complete simulation lifecycle, SIMULOS® establishes a technology-neutral methodology for ensuring that simulation-derived knowledge remains traceable to the object, purpose, representation, assumptions, models, inputs, scenarios, executions, outputs, correlations, uncertainties, limitations, and changing reality from which its legitimate meaning originates. SIMULOS® therefore enables simulation to become more powerful without allowing computational representation to become indistinguishable from Operational Reality, and provides the methodological infrastructure required for simulation-derived evidence to support engineering, science, AI, autonomy, safety, validation, mission planning, critical infrastructure, and future operational environments with explicitly governed meaning and reliance.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>